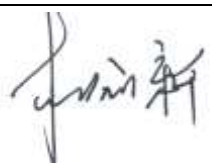
Product Specification

（Common Application）

Product Name: KEGM042602E02

Product Code: P426051

Customer
Approved by Customer
Approved Date:

Designed By	Checked By	Approved By	
		R&D	QA
			

CONTENT

REVISION RECORD **3**
1 APPLICATION FILED **4**
2 OVERVIEW **4**
3 FEATURES **4**
4 MECHANICAL DATA **4**
5 MECHANICAL DRAWING **5**
6 MODULE INTERFACE **6**
7 ABSOLUTE MAXIMUM RATINGS **7**
8 ELECTRICAL CHARACTERISTICS **7**
8.1 DC CHARACTERISTICS 7
8.2 PANEL DC CHARACTERISTICS (DRIVER IC INTERNAL REGULATORS) 8
8.3 OPTICAL SPECIFICATION 8
8.4 AC ELECTRICAL CHARACTERISTICS 9
9 FUNCTIONAL SPECIFICATION AND APPLICATION CIRCUIT **12**
9.1 POWER ON/POWER OFF SEQUENCE 12
9.2 TYPICAL APPLICATION CIRCUIT WITH SPI INTERFACE 14
9.3 DISPLAY CONTROL INSTRUCTION 15
9.4 RECOMMENDED SOFTWARE INITIALIZATION 15
10 PACKAGE SPECIFICATION **21**
11 RELIABILITY **22**
12 ILLUSTRATION OF EPD PRODUCT NAME **23**
13 OUTGOING QUALITY CONTROL SPECIFICATIONS **24**
13.1 SAMPLING METHOD 24
13.2 INSPECTION CONDITIONS 24
13.3 QUALITY ASSURANCE ZONES 24
13.4 INSPECTION STANDARD 25
14 HANDLING, SAFETY, AND ENVIRONMENT REQUIREMENTS **30**

REVISION RECORD

REV.	REVISION DESCRIPTION	REV. DATE	REMARK
Y01	Initial Release	2024-08-21	

1 Application Filed

Common Application

2 Overview

KEGM042602E02 is a reflective electrophoretic display module on an active matrix TFT substrate, The diagonal length of the active area is 4.26" and contains 480 x800 pixels. The panel is capable of displaying 1-bit black, white, red and yellow images depending on the associated lookup table used. The circuitry on the panel includes an integrated gate and source driver, timing controller, oscillator, DC-DC boost circuit, and memory to store the frame buffer and lookup tables, and additional circuitry to control VCOM and border settings.

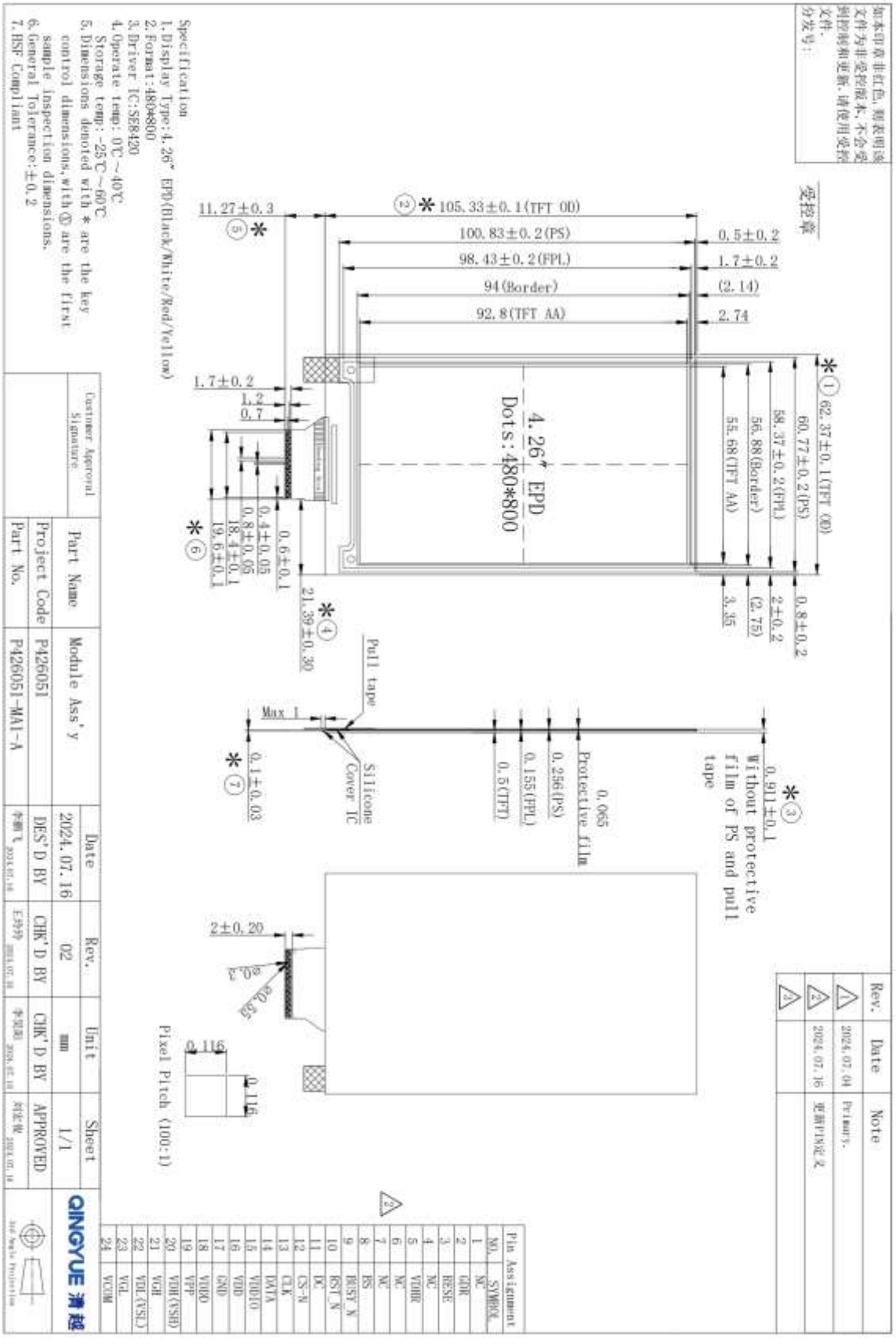
3 Features

- Ultra low power consumption
- Ultra wide viewing angle
- On chip display RAM
- Interface: 4-Wire SPI or 3-Wire SPI
- Wide range of operating temperature: 0℃ to 40℃
- Wide range of Storage temperature: -25℃ to 60℃
- High reflectance and contrast TFT electrophoretic.

4 Mechanical Data

NO.	ITEM	SPECIFICATION	UNIT
1	Dot Matrix	480(H)×800(V)	-
2	Screen Size	4.26	Inch
3	Active Area	55.68(H)×92.8(V)	mm
4	Pixel Pitch	0.116×0.116	mm
5	Pixels Per Inch	219	–
6	TFT Area	62.37(H)×105.33 (V)	mm
7	Outline Dimension	62.37(H)×116.6(V) ×0.911(D)	mm
8	Pixel Configuration	Square	-
9	Driver IC	SE8420	-
10	Module Weight	TBD±10%	g

5 Mechanical Drawing



6 Module Interface

PIN NO.	PIN NAME	DESCRIPTION
1	NC	No Connection
2	GDR	This pin is N-Channel MOSFET gate drive control pin.
3	RESE	Current Sense Input for the control loop
4	NC	No Connection
5	VDHR	This pin is Positive Source driving voltage, VSH2 connect a stabilizing capacitor between VSH2 and VSS in the application circuit.
6	NC	No Connection
7	NC	No Connection
8	BS	This pin is for selecting 3-wire(H active) or 4-wire(L active) SPI interface.
9	BUSY_N	This pin is Busy state output pin. When Busy is High, the operation of the chip should not be interrupted, and command should not be sent. For example, The chip would output Busy pin as High when - Outputting display waveform; or - Programming with OTP - Communicating with digital temperature sensor In the cascade mode, the BUSY pin of the slave chip should be left open.
10	RST_N	This pin is reset signal input (Active Low).
11	DC	This pin is Data/Command control pin connecting to the MCU
12	CS_N	This pin is the chip select input connecting to the MCU.
13	CLK	This pin is serial clock pin for interface.
14	DATA	This pin is serial data pin for interface.
15	VDDIO	Power input pin for the Interface. Connect to VCI in the application circuit.
16	VDD	Core logic power pin VDD can be regulated internally from VCI. A capacitor should be connected between VDD and VSS under all circumstances
17	GND	Ground
18	VDDD	Power input pin for the chip.
19	VPP	Power Supply for OTP Programming.
20	VDH(VSH)	This pin is Positive Source driving voltage, VSH1 Connect a stabilizing capacitor between VSH1 and VSS in the application circuit.
21	VGH	This pin is Positive Gate driving voltage. Connect a stabilizing capacitor between VGH and VSS in the application circuit.
22	VDL(VSL)	This pin is Negative Source driving voltage. Connect a stabilizing capacitor between VSL and VSS in the application circuit.
23	VGL	This pin is Negative Gate driving voltage. Connect a stabilizing capacitor between VGL and VSS in the application circuit.
24	VCOM	This pins is VCOM driving voltage Connect a stabilizing capacitor between VCOM and VSS in the application circuit.

7 Absolute Maximum Ratings

ITEM	SYMBOL	MIN	MAX	UNIT	REMARK
Logic supply voltage	VCI/VDD	-0.5	+4.0	V	-
I/O supply voltage	VDDIO	-0.5	+4.0	V	-
Logic Input voltage	V_{IN}	-0.5	VDDIO+0.5	V	-
Operating Temp.	Top	0	+40	℃	-
Storage Temp	Tstg	-25	+60	℃	-

Note (1): All of the voltages are on the basis of "VSS = 0V".

Note (2): Maximum ratings are those values beyond which damages to the device may occur. Functional operation should be restricted to the limits in the Panel DC Characteristics tables.

8 Electrical Characteristics

8.1 DC Characteristics

The following specifications apply for: VSS=0V, VCI=3.3V, TOPR =23ºC.

Parameter	Symbol	Condition	Applicable pin	Min.	Typ.	Max.	Unit
Supply Voltage	VCI	-	VCI	2.3	3.0	3.6	V
IO supply voltage	VDDIO	-	-	2.3	3.0	3.6	V
Digital voltage	VDD	-	-	2.3	3.0	3.6	V
OTP Program voltage	VPP	-	-	8.1	8.3	8.5	V
High level input voltage	V_{IH}	SDA SCL, CSB, D/C, RES, BS1	-	$0.8V_{VDDIO}$	-	-	V
Low level input voltage	V_{IL}	SDA SCL, CSB, D/C, RES, BS1	-	-	-	$0.2V_{VDDIO}$	V
High level output voltage	V_{OH}	SDA, BUSY	-	$0.8V_{VDDIO}$	-	-	V
Low level output voltage	V_{OL}	SDA, BUSY	-	-	-	$0.2V_{VDDIO}$	V
Typical power panel	P_{TYP}	-	-	-	16.5	-	mW
Standby power panel	P_{STPY}	-	-	-	0.0066	-	mW
Typical operating current(white state)	Iopr_VCI	-	-	-	15.7	-	mA
Image update time	-	23 ºC	-	-	5	-	sec
Sleep mode current	Islp_VCI	Ta=23℃ VDDIO=VDD=3V VGH=20,VGL=-20 VDH1=15,VDL=-	VCI	-	40	70	uA
Digital deep sleep current	I_{VDD}	Pattern no update, OSC off, VDD2/VDD3=3.0V, Ta=23 ℃	VCI	-	2	6	uA

Note: The power input must be kept in a stable value; ripple and noise are not allowed.

8.2 Panel DC Characteristics (Driver IC Internal Regulators)

The following specifications apply for: VSS=0V, VCI=3.3V, TOPR =23ºC.

Parameter	Symbol	Condition	Applicable pin	Min.	Typ.	Max.	Unit
VCOM output voltage	VCOM	-	VCOM	-	-2.0	-	V
Positive Source output voltage	V_{SH}	-	S_0~S_{479}	-	+15	-	V
Negative Source output voltage	V_{SL}	-	S_0~S_{479}	-	-15	-	V
Positive gate output voltage	Vgh	-	G_0~G_{799}	19.5	+20	20.5	V
Negative gate output voltage	Vgl	-	G_0~G_{799}	-	-20	-	V

8.3 Optical Specification

Measurements are made with that the illumination is under an angle of 45 degree, the detection is perpendicular unless otherwise specified.

Symbol	Parameter	Conditions	Min	Typ.	Max	Units	Notes
R	White Reflectivity	White	28		-	%	8-1
CR	Contrast Ratio	indoor	20	-	-	-	8-2
T update	Image update time	23 °C	-	TBD	-	sec	-
Tlife	Life	Topr	-	1000000 times or 5years	-	-	-

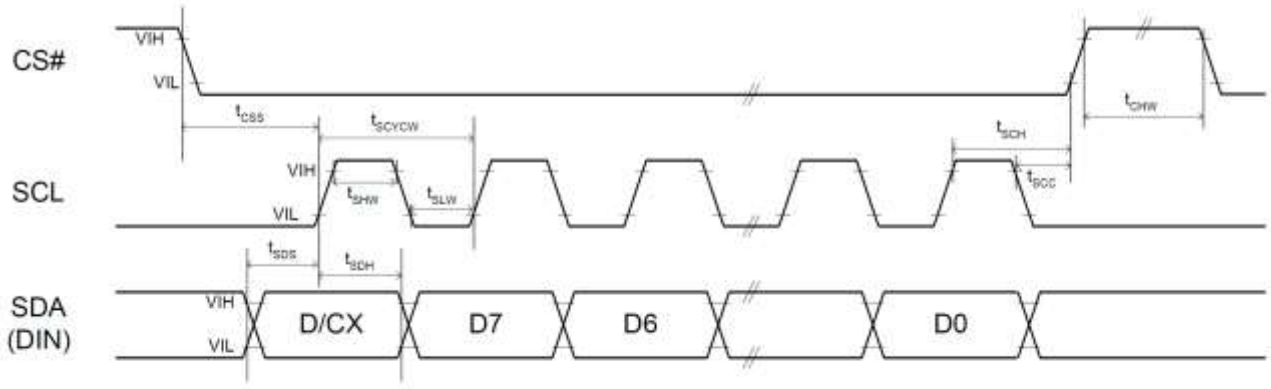
Notes1: Luminance meter: Eye-One Pro Spectrophotometer.

Notes2:CR=Surface Reflectance with all white pixel/Surface Reflectance with all black pixels.

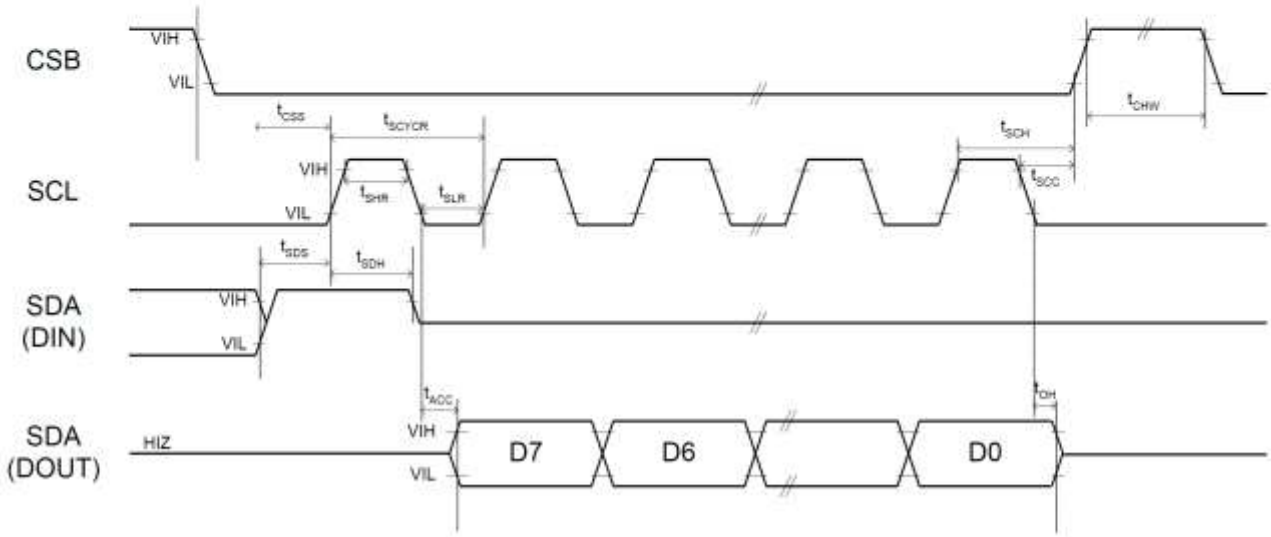
8.4 AC Electrical Characteristics

(1) Serial Peripheral Interface

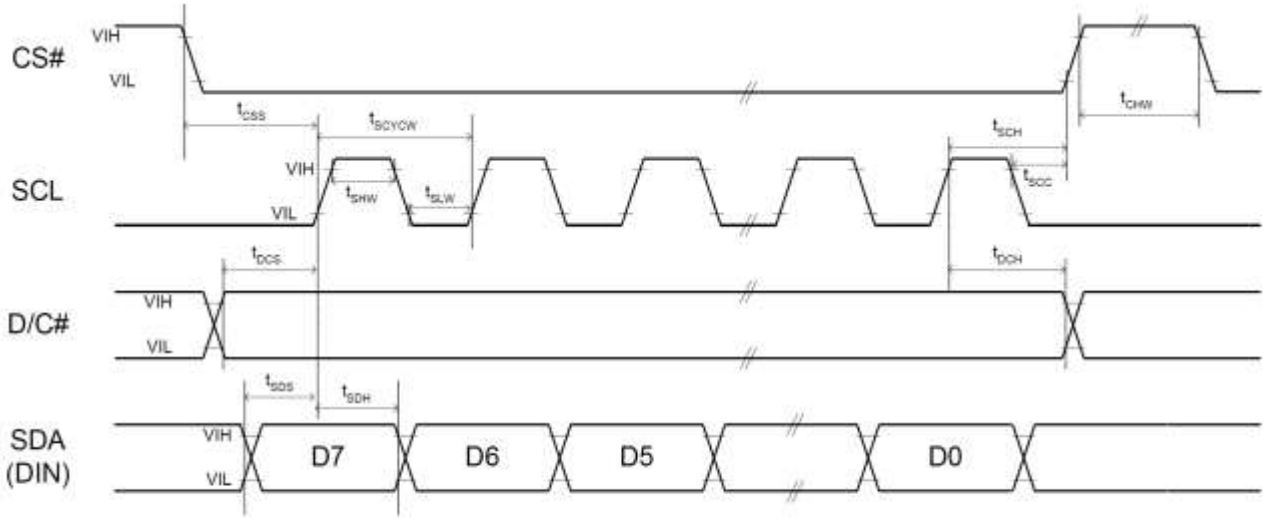
Symbol	Parameter	Min	Typ	Max	Unit
t_{CSS}	CSB select setup time	TBD			ns
t_{SCH}	CSB select hold time	TBD			ns
t_{SCC}	CSB deselect setup time	TBD			ns
t_{CHW}	CSB deselect hold time	TBD			ns
t_{SCYCW}	Serial clock cycle (Write)	TBD			ns
t_{SHW}	SCL "H" pulse width (Write)	TBD			ns
t_{SLW}	SCL "L" pulse width (Write)	TBD			ns
t_{SCYCL}	Serial clock cycle (Read)	TBD			ns
t_{SHR}	SCL "H" pulse width (Read)	TBD			ns
t_{SLR}	SCL "L" pulse width (Read)	TBD			ns
t_{SDS}	Data setup time	TBD			ns
t_{SDH}	Data hold time	TBD			ns
t_{ACC}	Access time			TBD	ns
t_{OH}	Output disable time	TBD			ns



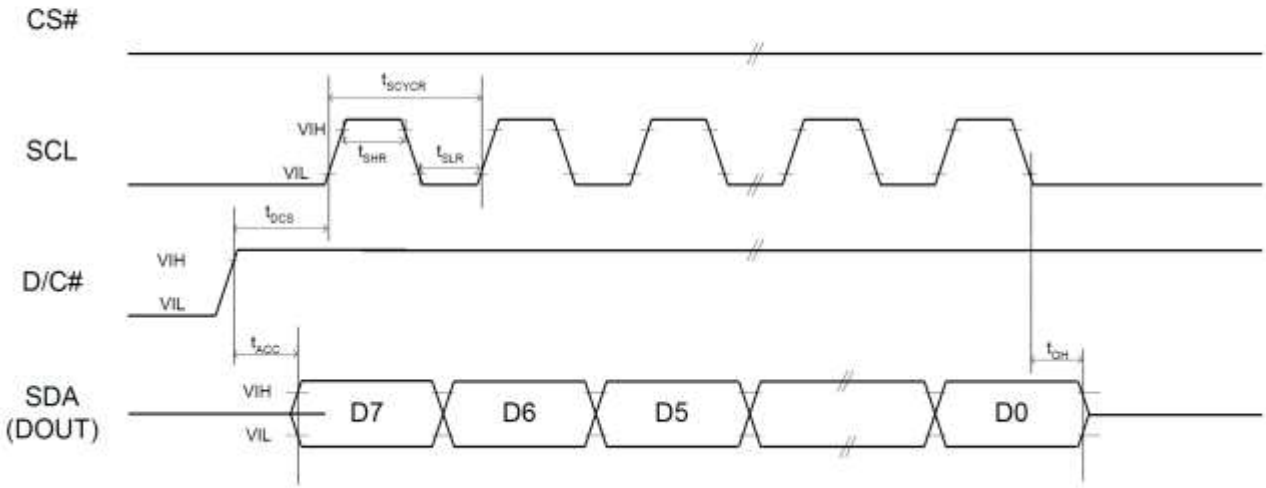
3 pin serial interface characteristics (write mode)



3 pin serial interface characteristics (read mode)



4 pin serial interface characteristics (write mode)

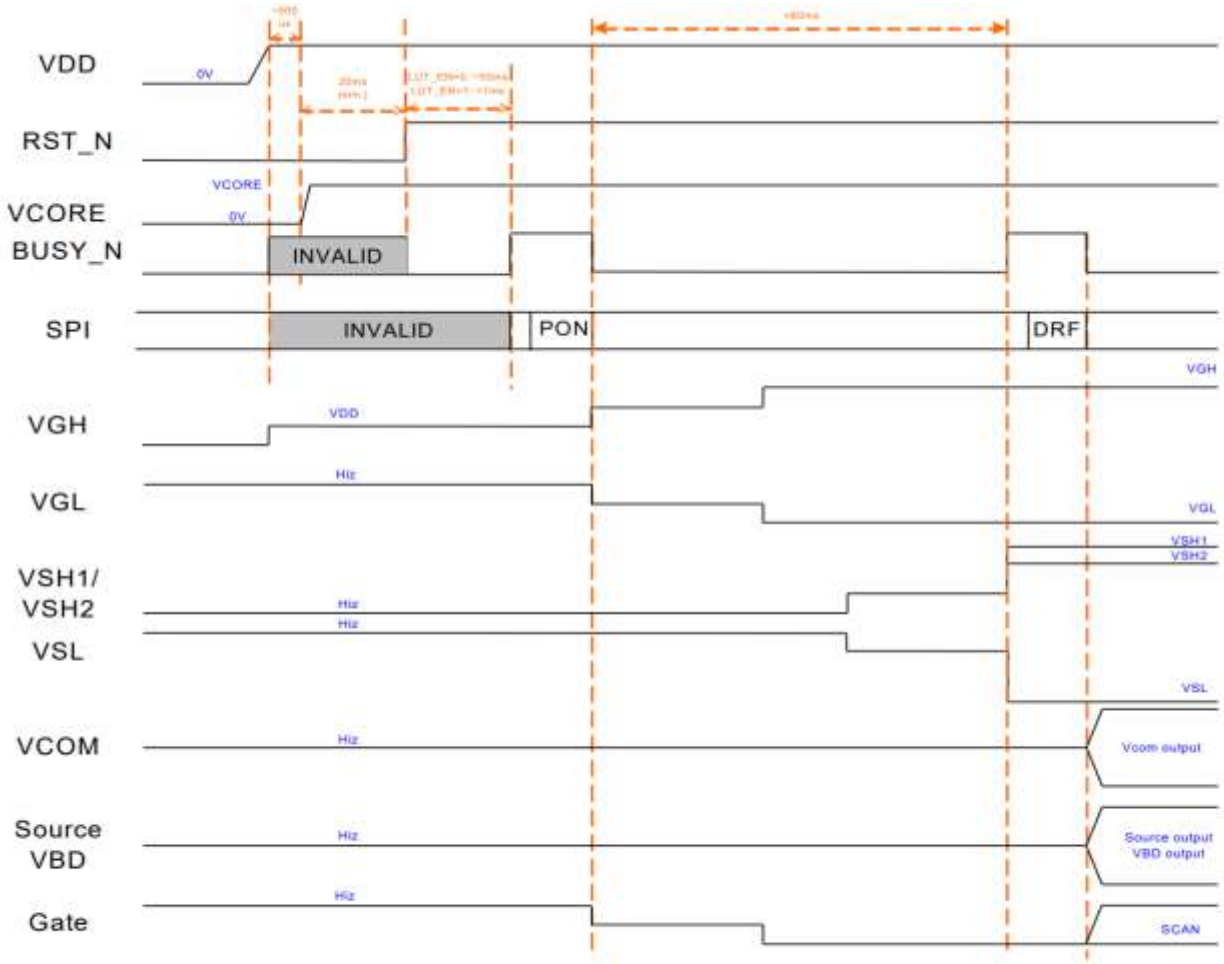


4 pin serial interface characteristics (read mode)

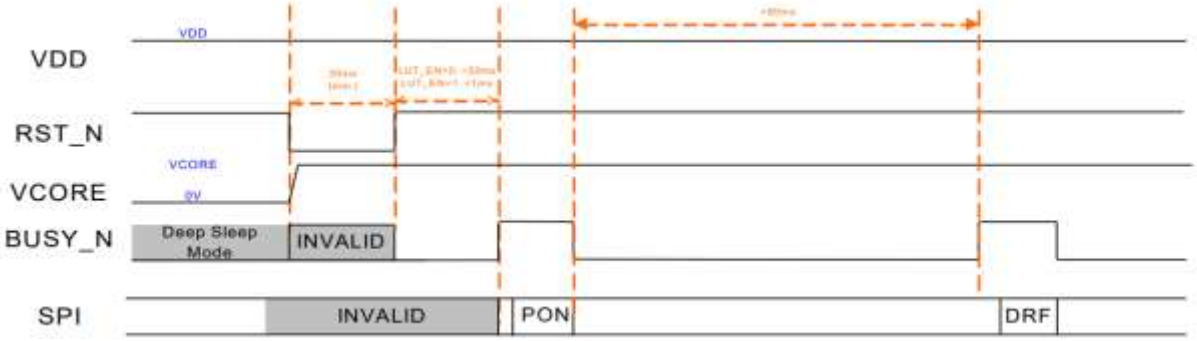
9 Functional Specification and Application Circuit

9.1 Power ON/Power OFF Sequence

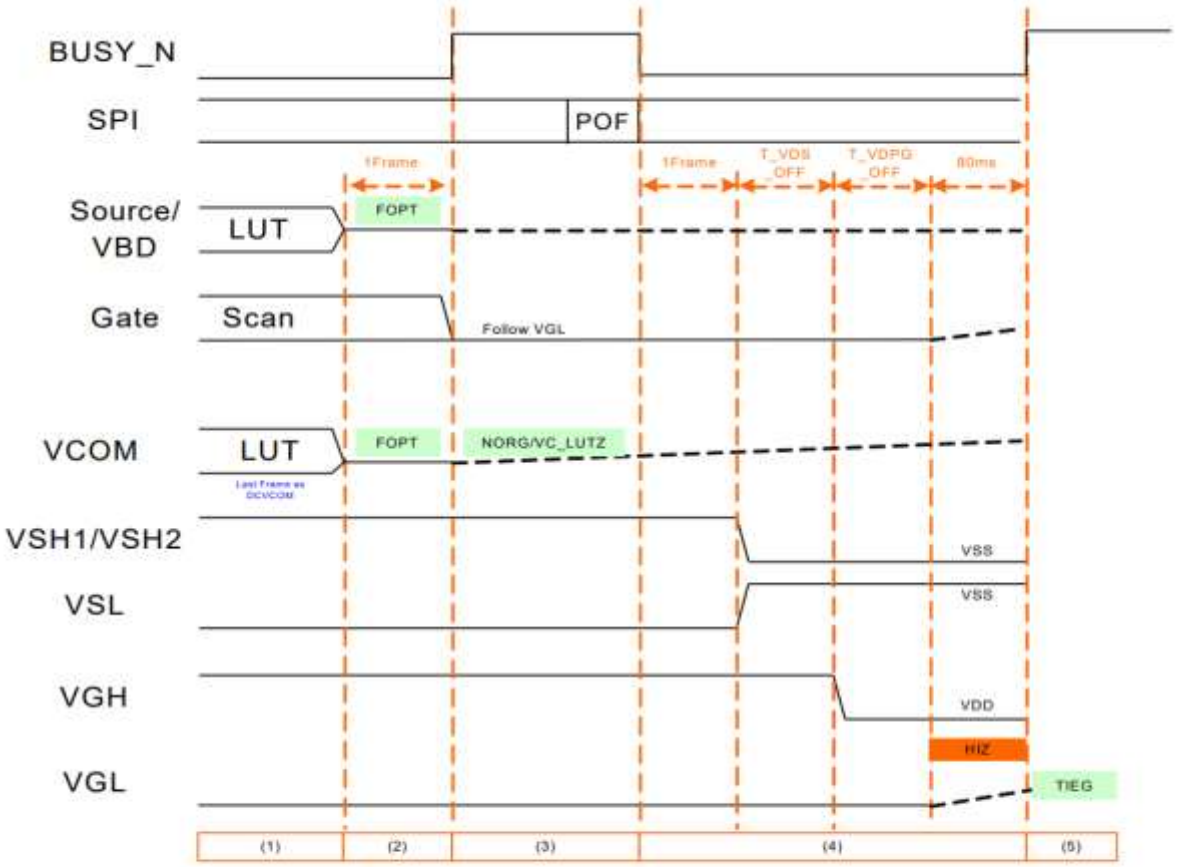
Power on Sequence from VDD power on:



Display on sequence from deep sleep mode:

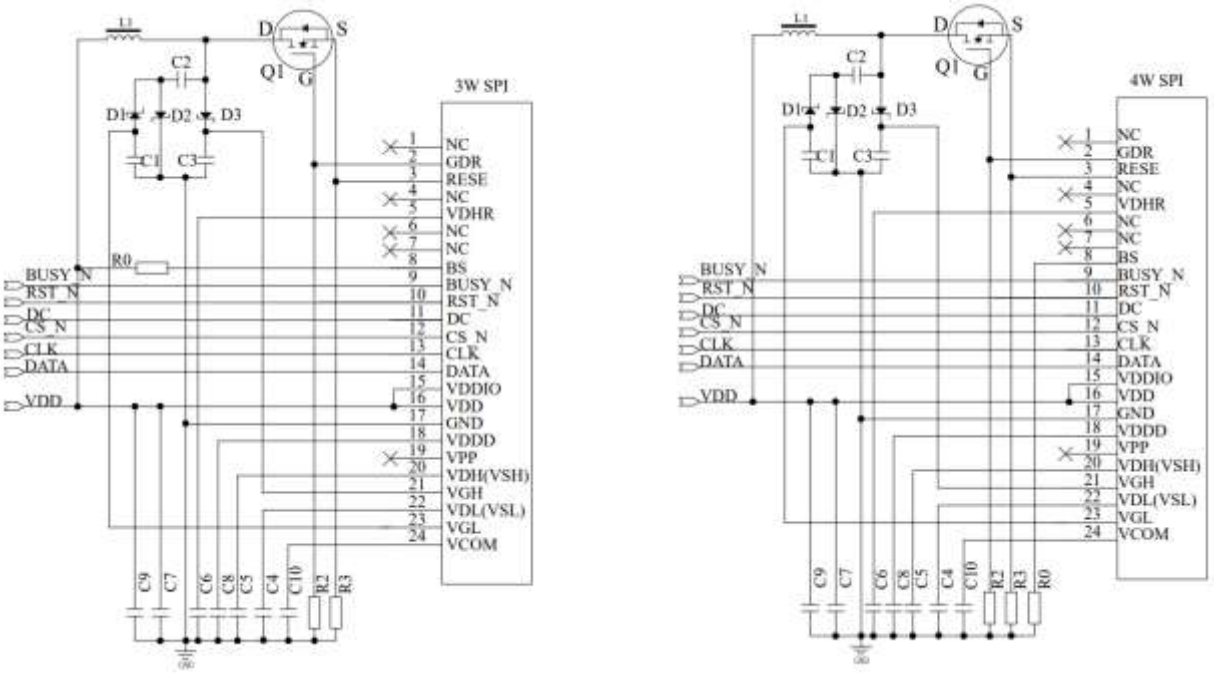


Power off Sequence:



9.2 Typical Application Circuit with SPI Interface

(1). Schematic of application circuit



Component list for application circuit:

Part Name	Description	Reference Part/Requirement
C7,C9	Chip Ceramic Capacitor	4.7uF 6Vor25V X5R/X7R ±10%
C8	Chip Ceramic Capacitor	1uF 25V X5R/X7R ±10%
C1-C6	Chip Ceramic Capacitor	4.7uF 25V X5R/X7R ±10%
C10	Chip Ceramic Capacitor	1uF 25V X7R ±10%
R3	Chip Resistor	2.2ohm 1% 1/20W ±10%
R0,R2	Chip Resistor	10 K ohm 1% 1/20W ±10%
D1-D3	Schottky Rectifier Diode	OnSemi MBR0530 (V_R > 25V, I_F > 500mA, I_R < 1mA @ V_R=15V, Ta=100℃)
Q1	N-Channel MOSFET	Vishay Si1308EDL (V_{DS} > 25V, I_D > 500mA, V_{GS}(th) < 1.5V, C_{iss} < 200pF, R_{DS}(on) < 400mΩ)
L1	Chip Inductor	47uH±30% Io≤1A

9.3 Display Control Instruction

Refer to SE8420 IC Specification.

9.4 Recommended Software Initialization

In order to ensure the reliability and stability of the module, the module must initialized use the following code, Malfunctioning of the module may occur and the reliability of the module may deteriorate if the module is used beyond the initialize code.

```
void driver_delay_xms(unsigned long xms)
{
   unsigned long i = 0 , j=0;

   for(j=0;j<xms;j++)
    {
      for(i=0; i<256; i++);
   }
}
void SPI_Delay(unsigned int xrate)
{
    unsigned char i;
    while(xrate)
    {
          for(i=0;i<EPD_W21_SPI_SPEED;i++);
          xrate--;
    }
}
void EPD_W21_REST(void)
{
    EPD_W21_RST_0;
    driver_delay_xms(100);
    EPD_W21_RST_1;
    driver_delay_xms(10);

}
void Busy_chkstatus(void)
{
    while(1)
    {
```

```
            if(isEPD_W21_BUSY==1) //busy
            break;
    }
    driver_delay_xms(10);
}
void SPI_Write(unsigned char value)
{
  unsigned char i;
    SPI_Delay(1);
    for(i=0; i<8; i++)
    {
            EPD_W21_CLK_0;
            SPI_Delay(1);
            if(value & 0x80)
            EPD_W21_MOSI_1;
            else
            EPD_W21_MOSI_0;
            value = (value << 1);
            SPI_Delay(1);
            driver_delay_us(1);
            EPD_W21_CLK_1;
            SPI_Delay(1);
    }
}
void EPD_W21_WriteCMD(unsigned char command)
{
    SDA_OUT();
    SPI_Delay(1);
    EPD_W21_CS_0;
    EPD_W21_DC_0;              // command write
    SPI_Write(command);
    EPD_W21_CS_1;
}
void EPD_W21_WriteDATA(unsigned char data)
{
    SDA_OUT();
    SPI_Delay(1);
```

```
    EPD_W21_CS_0;
    EPD_W21_DC_1;              // command write
    SPI_Write(data);
    EPD_W21_CS_1;
}

void EPD_W21_Init(void)
{
        EPD_W21_REST();  //硬件复位
        Busy_chkstatus();

        EPD_W21_WriteCMD(0x00);  //0x00
        EPD_W21_WriteDATA(0x0F);  // bit5=1 ModeB
        EPD_W21_WriteDATA(0x29);
        Busy_chkstatus();

    EPD_W21_WriteCMD(0x06);//0x06
    EPD_W21_WriteDATA(0x0F);
    EPD_W21_WriteDATA(0x8B);
    EPD_W21_WriteDATA(0x93);
    EPD_W21_WriteDATA(0xA4);

    EPD_W21_WriteCMD(0x61);
    EPD_W21_WriteDATA(0x03);// Source Line
    EPD_W21_WriteDATA(0x20);
    EPD_W21_WriteDATA(0x02);// Gate Line
    EPD_W21_WriteDATA(0xA8);
    Busy_chkstatus();

    EPD_W21_WriteCMD(0x50);
    EPD_W21_WriteDATA(0x37);

    EPD_W21_WriteCMD(0x30);
    EPD_W21_WriteDATA(0x08);

    EPD_W21_WriteCMD(0x62);
    EPD_W21_WriteDATA(0x76);
```

```
    EPD_W21_WriteDATA(0x76);
    EPD_W21_WriteDATA(0x76);
    EPD_W21_WriteDATA(0x5A);
    EPD_W21_WriteDATA(0x9D);
    EPD_W21_WriteDATA(0x8A);
    EPD_W21_WriteDATA(0x76);
    EPD_W21_WriteDATA(0x62);

        EPD_W21_WriteCMD(0xE3);//0xE3
        EPD_W21_WriteDATA(0x22);

    EPD_W21_WriteCMD(0xE0);
    EPD_W21_WriteDATA(0x10);

    EPD_W21_WriteCMD(0xE7);
    EPD_W21_WriteDATA(0xA4);

    EPD_W21_WriteCMD(0xE9);
    EPD_W21_WriteDATA(0x01);


}

void EPD_ALLScreen(unsigned char color)
{
    unsigned int i,j;

    EPD_W21_WriteCMD(0x10);                 //开始传输图像
    for(j=0;j<800*480/4;j++)
    {
        EPD_W21_WriteDATA(color1);
    }

}
```

```
void EPD_Update_and_Sleep(void)
{
    EPD_W21_WriteCMD(0X04);//power on
    delay_ms(100);
    Busy_chkstatus();

    EPD_W21_WriteCMD(0X12);//update
    EPD_W21_WriteDATA(0x00);
    delay_ms(100);
    Busy_chkstatus();

    EPD_W21_WriteCMD(0X02);//power off
    EPD_W21_WriteDATA(0x00);
    Busy_chkstatus();

    EPD_W21_WriteCMD(0X07);//deepsleep
    EPD_W21_WriteDATA(0xA5);
}




int main(void)
{

    RCC_Configuration();
    GPIO_Configuration();
    RS232_Init(9600);
    NVIC_Configuration();
    KEYGPIO_Init();




    EPD_W21_Init();
    EPD_ALLScreen(0x00);//black
    EPD_Update_and_Sleep();
```

```
    driver_delay_xms(5000);

    EPD_W21_Init();
    EPD_ALLScreen(0x55);//white
    EPD_Update_and_Sleep();
    driver_delay_xms(5000);

    EPD_W21_Init();
    EPD_ALLScreen(0xAA);//Yellow
    EPD_Update_and_Sleep();
    driver_delay_xms(5000);

    EPD_W21_Init();
    EPD_ALLScreen(0xFF);//Red
    EPD_Update_and_Sleep();
    driver_delay_xms(5000);



}
```

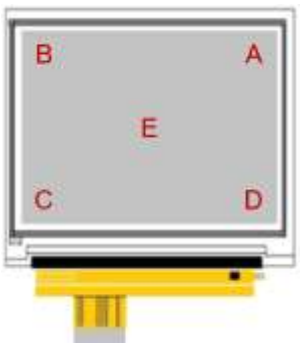
10 Package Specification

NOTE:1、The inner cartoster carton must be sealed with adhesive tape.
2、Fill up the gap with empty tray.
3、If the customer has special needs with the RoHS making, the inner carton and master carton need adhesive new RoHS marking at◎.
4、Packaging materials are not recommended for recycling.

11 Reliability

NO	Test items	Test condition	QUANTITY
1	Low-Temperature Storage	T = -25℃ , low temperature film T= -30℃; White screen state, for 500h.	5pcs
2	Low-Temperature Operation	T = 0°C, 500 h; Put the product into the experimental procedure, run it in the temperature box, and check it every 24 hours.	5pcs
3	High-Temperature Operation	T = 40°C, RH = 35%, 500 h; Put the product into the experimental procedure, run it in the temperature box, and check it every 24 hours.	5pcs
4	High-Temperature Storage	T=60ºC，RH=35%; White screen state, for 500h.	5pcs
5	High-Temperature/ High- humidity	T=40ºC，RH=90%; White screen state, for 500h.	5pcs
6	Temperature Cycle	1 cycle:[-25°C 30min]→[+40 °C 30 min]; 100 cycles.	5pcs
7	High-Temperature/ High- humidity Storage	T=50ºC，RH=90%; White screen state, for 500h.	5pcs
8	UV exposure Resistance	765W/m² for 168hrs,T = 40°C, RH=35%;	5pcs
9	ESD Contact discharge	Air±8 KV, Contact±6 KV, Test 5 point; Each point discharge 10 times. Time interval is not less than 1 second.	5pcs

ESD test location



Test and measurement conditions

After the end of the experiment, the sample was taken out of the temperature chamber, and stood at room temperature for 1h, and then the sample was inspected for appearance, function and optical inspection.

Criteria for qualification (pass the test if all qualified) :

(1) The product can be normal refresh.

(2) There are no new point defects or line defects in the display screen.

(3) No discoloration, blurred handwriting and barcode can be read on the complex screen.

12 Illustration of EPD Product Name

□ □ □ □ □□□□ □□ □ □□

第 12~13 码代表流水码
01~99

第 11 码代表显示颜色
M: 双色
T: 黑白红三色 Y: 黑白黄三色
F: 全彩 E: E5 P: Prism3

第 9-10 码代表屏体流水码
01~99

第 5-8 码代表产品对角线尺寸
0213: 2.13″

第 4 码代表产品类型
M:模组 P:屏体

第 3 码代表显示类型
G:图形；C 字符；S：断码；M：混合

第 2 码代表产品类别
E:电子纸

第 1 码代表公司
K:清越

13 Outgoing Quality Control Specifications

13.1 Sampling Method

(1) GB/T 2828.1, inspection level II, normal inspection, single sample inspection

(2) AQL: Major 0.4; Minor 0.65

13.2 Inspection Conditions

The environmental conditions for test and measurement are performed as follows.

Temperature: 23±3°C

Humidity: 55±15%R.H

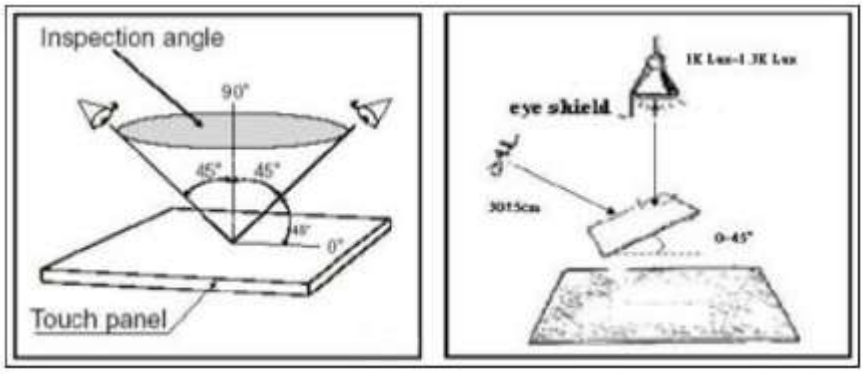
Inspection of illuminance: 800~1500Lux

Inspection time: signal face 5S-10S

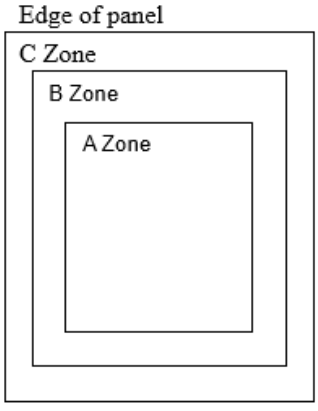
Distance between the Panel & Eyes: 30±10cm

Viewing angle from the vertical in each direction: ±45°

(See the sketch below)



13.3 Quality Assurance Zones



Zone A：Active Area

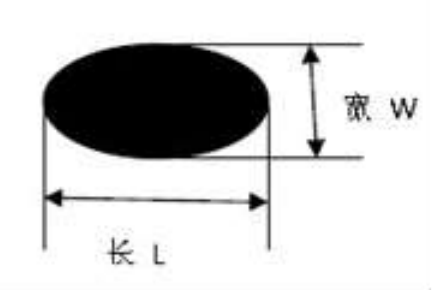
Zone B: Black Frame Area

Zone C: Outside Black Frame Area

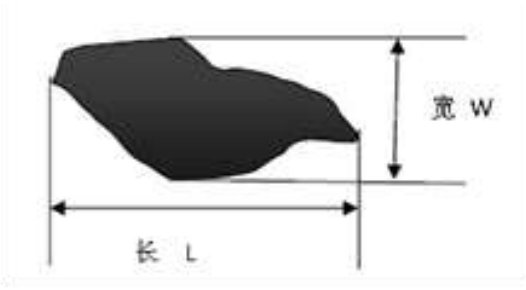
13.4 Inspection Standard

Defects Definition of Φ&L&W (Unit: mm)

13.4.1 Dot defects:

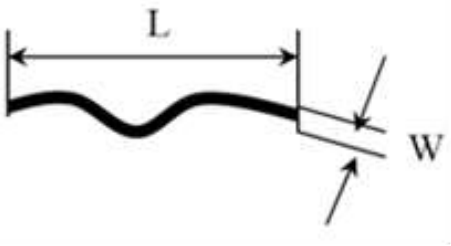


ΦD=Max(L,W)

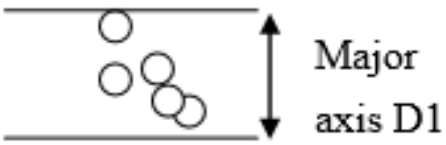


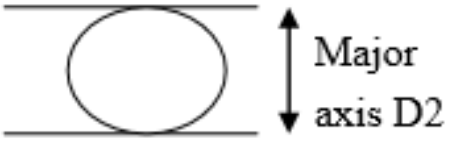
ΦD=Max(L,W)

13.4.2 line defect:

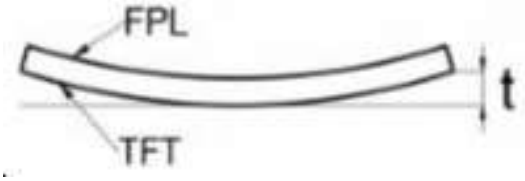


13.4.3 Small bubble aggregation and large bubble definition:

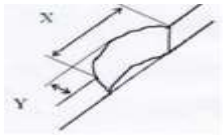
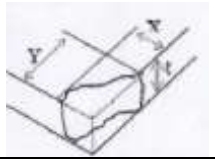




13.4.4 TFT warpage:

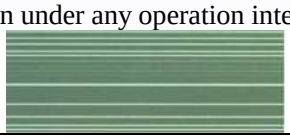


I. Appearance Defects

NO.	ITEM	CRITERIA	Acceptable range	Method	Defect level	Area
1	Spotty (black spots, white spots, foreign bodies, air bubbles, bumps)	D≦0.25mm	Ignore	Film Card	Minor	Zone A
		0.25mm < D≦0.5 mm, Distance≥5mm	N≦4			
		D>0.5 mm	N=0			
		0.1mm< D≦0.25 mm(Dense point)	N≦3/cm2			
2	POOR LINEAR SHAPE (foreign body, glass scratch)	L≦2mm, W≦0.1mm	Ignore	Film Card	Minor	Zone A
		2mm<L≦8mm, 0.1<W≦0.5mm	N≦4			
		L>8mm, W>0.5mm Note: FPL Lacerations are not allowed	N=0			
	steel pit	Long strip pits are not allowed	N=0	Sight Check	Minor	Zone A
3	Glass Crack	Extensional cracks are not allowed 	N=0	Sight Check	Major	Zone B,C
4	Edge breakage	X≤3mm,Y≤0.5mm, It does not affect the electrode 	N≦2	Sight Check/ Microscope	Minor	Zone C
5	Chip Package Chip Off	X≤2mm，Y≤2mm, It does not affect the electrode(FPC edge) X≤1mm，Y≤1mm, It does not affect the electrode((Not FPC edge) 	N≦2	Sight Check/ Microscope	Minor	Zone C
6	Dirt	No dirt (finger print, dust, residual glue, etc.)	Ignore	Sight Check	Minor	Zone A,B
7	Silicone	The maximum diameter of a single bubble cannot exceed 2mm	N≤2	Sight Check/ Film card	Minor	Zone C
		Crack is not allowed and there are no visible impurities in the glue of the lead part	N=0			
		The adhesive must completely cover the ACF, lead area and IC and should be applied evenly	N=0			

NO.	ITEM	CRITERIA	Acceptable range	Method	Defect level	Area
		No glue leakage, no obvious lack of glue in the lead area	N=0			
		Glue height exceeds PS surface	N=0			
		FPC Front overflow glue width>0.5mm or Back side overflow glue width>1mm	N=0			
8	Edge Sealing Adhesive	No glue leakage	N=0	Sight Check/ Film card	Major	Zone C
		The height of sealant exceeds PS surface	N=0		Minor	
		The edge sealing adhesive shall not leak the TFT glass substrate	N=0			
		Judging Ok of water-blocking area ≥0.7 mm of PS edge sealant	N=0		Minor	
9	Protective film	Foreign body in protective film	N=0	Sight Check	Minor	Zone A
		The protective film punctures and injures FPL	N=0			
10	Pull Tape	Attachment position is wrong Cannot tear up the protective film	N=0	Sight Check	Minor	Zone C
11	FPC	FPC has break, scratch, gold finger stripping or oxidation, dirty, residual glue	N=0	Sight Check	Major	Zone C
12	Glass edge bulge	X≤3mm，Y≤0.3mm	N≦1	Sight Check	Minor	Zone C
13	Warping	t＞1mm （3.5inch below） t＞3mm （3.5inch above）	N=0	Plug Gage	Minor	Zone C
14	Chromatism	Color difference in silver paste area (Not in Zone A)	Ignore	Sight Check	Minor	Zone C
		FPL Peeling occurs, chromatic aberration occurs	N=0	Sight Check	Major	Zone A,B
		The color difference of side loss of FPL in zone B ≥1/2 width	N=0	Sight Check	Major	Zone A,B
15	Silver pulp point	FPL and TFT substrate conduction, silver point <1.0mm (Both single silver point and double silver point should meet this specification)	N=0	Film card	Major	Zone C
16	Inkjet code	The ink jet printing font is clear, identifiable, and cannot be missing	N=0	Sight Check	Minor	Zone C

II. Displaying Defects

NO.	ITEM	CRITERIA	Acceptable range	Method	Defect level	Area
1	Poor DOT SHAPE (black, white, group White)	D≦0.25mm	Ignore	Film Card	Major	Zone A
		0.25mm < D≦0.5mm, Distance≥5mm	N≦4			
		D>0.5mm	N=0			
		0.1mm< D≦0.25 mm(Dense point)	N≦3/cm2			
2	Line defects	White or black lines running through the entire screen under any operation interface 	N=0	Sight Check	Major	Zone A
3	ghost	Ghosts appear only during screen switching	Ignore	Sight Check	Major	Zone A
4	Flash Point	Flash point occurs during screen switching only	Ignore	Sight Check	Major	Zone A
5	Flash Line	Flash line occurs during screen switching only	N=0	Sight Check	Major	Zone A
6	Display screen error	Unable to display a fixed screen correctly	N=0	Sight Check	Major	Zone A
7	Display abnormal	No display, The red matrix darkens, Note fuzzy, bar code can not be scanned,	N=0	Sight Check	Major	Zone A
8	Residual image	Residual Image Inspection (visual, final judgment reference optical specification)	N=0	Sight Check	Major	Zone A
9	Mura Anomaly	White/gray. Mura doesn’t allow it	N=0	Sight Check	Major	Zone A

Ⅲ. Identification and packaging inspection

NO.	ITEM	CRITERIA	Method	Defect level
1	Package	(1). The products are completely placed in the anti-static tray without overlapping. (2). Products with different models cannot be mixed in one internal packaging bag. (3) There is a desiccant in the packaging bag, with good internal packaging and no expansion of the packaging bag. (4) The Tray model, quantity and way used for packaging meet the requirements of product specifications.	Sight Check	Minor
2	Inner and outer packing	(1) No obvious deformation, damage, dampness or dirt on the packing case; (2) The type, quantity and method of the packing case used shall meet the requirements of the product specification. (3) There is no font or unclear design in the outer packing box.	Sight Check	Minor
3	Labels for inner and outer cases	(1). Any unnecessary marks or marks are not allowed to exist; (2). The label information such as model, specification, quantity, weight, material number, month label and environmental protection label should be clear and correct, which should be in line with product specifications or marked according to customer requirements.	Sight Check	Minor

14 Handling, Safety, and Environment Requirements

Warning
The display glass may break when it is dropped or bumped on a hard surface. Handle with care. Should the display break, do not touch the electrophoretic material. In case of contact with electrophoretic material, wash

Caution
The display module should not be exposed to harmful gases, such as acid and alkali gases, which corrode electronic components. Disassembling the display module.
Disassembling the display module can cause permanent damage and invalidates the warranty agreements.

Observe general precautions that are common to handling delicate electronic components. The glass can break and front surfaces can easily be damaged. Moreover the display is sensitive to static electricality and other rough environmental conditions.

Data sheet status	
Product specification	This data sheet contains final product specifications.
Limiting values	
Limiting values given are in accordance with the Absolute Maximum Rating System (IEC 134). Stress above one or more of the limiting values may cause permanent damage to the device. These are stress ratings only and operation of the device at these or at any other conditions above those given in the Characteristics sections of the specification is not implied. Exposure to limiting values for extended periods may affect device reliability.	
Application information	
Where application information is given, it is advisory and does not form part of the specification.	

Storage conditions
Under the conditions of temperature 10~30℃ and humidity 30~70%, The storage time was 12 months.